# PLANT HEALTH SCANNER APP

## WORKING EXPLANATION-FOR ANDROID APPLICATION

**DOCUMENT PURPOSE**: A simple explanation of how the completed system is structured and how its main parts work together.

**Application type:** Mobile application for Android and iOS, built with Flutter.

**Core technology:** Flutter (Dart), TensorFlow/Keras (MobileNetV3-Small CNN), TensorFlow Lite, Firebase Authentication, EmailJS, local JSON storage on the phone.

## 1.ProjectOverview:

The AI-Based Plant Health Scanner is designed to help farmers and home growers check the health of a crop leaf quickly. Instead of depending only on visual checking, the app uses a Convolutional Neural Network (CNN) to identify the crop from a leaf photo, decide whether the leaf is healthy or diseased, name the disease, and show a confidence score with a care tip. The system covers four crops (tomato, potato, corn and pepper) with 19 classes of healthy and diseased leaves. The AI model runs directly on the phone, so scanning works without internet. Every scan is saved in a history, and the result can be emailed to the signed-in user. The project is delivered as one Flutter mobile application that runs on Android and iOS.

## 2. Overall System Architecture

**1. Presentation layer:** The Flutter app shows the login, camera, result and history screens, and accepts photos from the camera or gallery.

**2. Application layer:** The app logic validates the photo, calls the AI engine, applies the confidence rule, saves the scan and sends the result email.

**3. AI layer:** The MobileNetV3-Small CNN, converted to TensorFlow Lite, classifies the leaf image on the phone.

**4. Data layer:** Scan history and photos are stored on the phone. Firebase keeps the user accounts, and EmailJS delivers the result emails.

**Basic request flow:** User → Camera or gallery → Image pre-processing → CNN prediction (TFLite) → Confidence check → Result screen → History save → Email to the signed-in user. The scan itself runs on the phone, so it does not need internet. Only sign-in and the result email use the network.

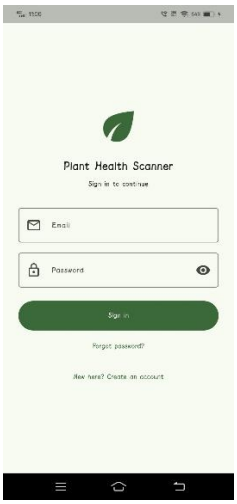
## 3. How the Mobile Application Works

A user opens the app, signs in and points the camera at a leaf. When the user takes a photo or picks one from the gallery, the app resizes the image, runs the AI model on the phone and shows the result. The app then saves the scan in the history and emails the result to the user.

**Typical app flow:**

- Open the app on the phone.
- Create account and after Sign in with the Email id and password.
- Point the camera at a crop leaf and tap scan, or choose a photo from the gallery.
- View the crop name, health status, confidence score and care tip.
- Receive the same result by email on the signed-in account.
- Open the history screen to review, delete or clear past scans.
- Sign out from the camera screen when finished.

## 4. Login and User Access



The phone login screen is the entry point for users. The user enters an email id with the password, and when the user enters the correct password, Firebase signs the user in.

## 5. How the AI Engine Works

**Image input:** The photo from the camera or gallery is read and resized to the model input size.

**Pre-processing:** Raw pixel values from 0 to 255 are given to the model. Its built-in preprocessing layer handles the scaling.

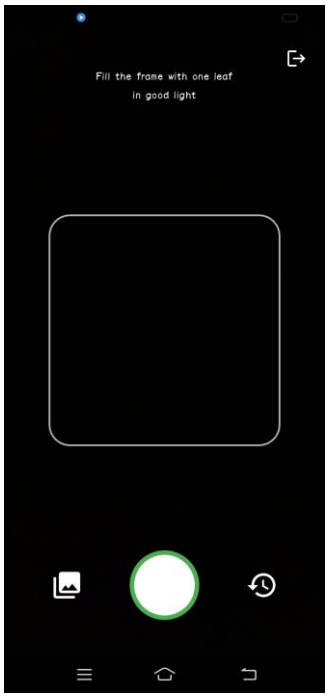
**CNN classification:** The trained MobileNetV3-Small network, built with TensorFlow/Keras and converted to TensorFlow Lite (float16), produces a probability for each of the 19 crop and disease classes.

**Confidence check:** The app takes the highest probability. A result is shown only when the confidence is at least 0.80.

**Result output:** The app shows the crop, the health condition or disease, the confidence score and a care tip.

## 6.How the Main Modules Work

### Camera and Gallery Scan

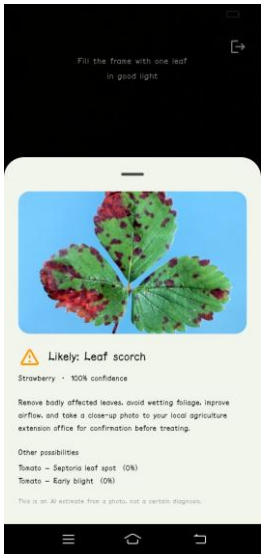


Shows a live camera preview and lets the user capture a leaf photo or pick one from the gallery.

**Leaf Classification (AI Detection)**

The classifier module loads the TensorFlow Lite model, runs it on the photo and returns the top predictions with their confidence.

**Result Screen**

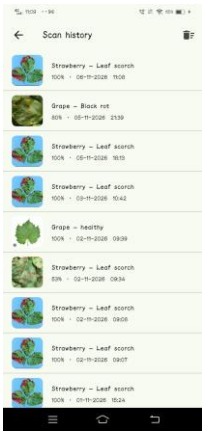



Shows the crop, healthy or diseased status, disease name, confidence score and care tip.

**Care Tips**

Each class is linked to short care advice that is shown with the result.

**Scan History and Records**

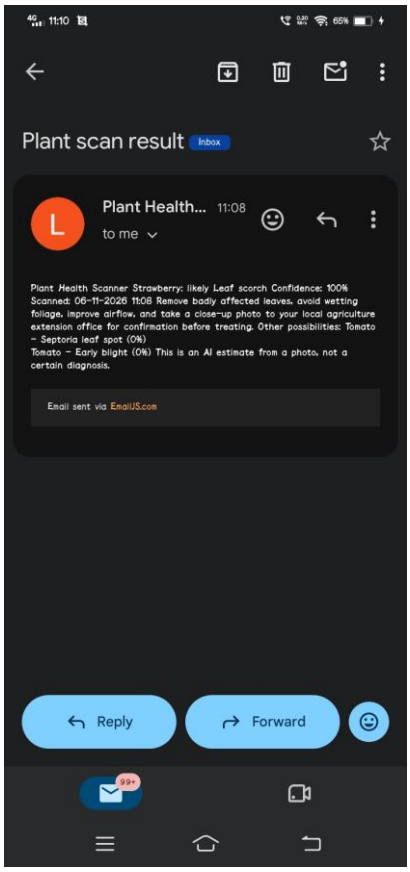



Stores up to 100 scans with their photos and results on the phone. Users can view past scans, swipe to delete one, or clear all

**Login and User Access**

Email id and password through Firebase Authentication. The scanner opens only for a signed-in user.

**Email Result**




After each scan, the result is sent to the signed-in user's email address through Email JS.